



Drones as back-up dancers at a stage performance at New York's Madison Square Garden
(Credit: www.rollingstone.com)

beaming columns of colour through fake smoke. Swarms of artificially-intelligent drones are now being used for stage performances. In 2017, Verity Studio's Lucie drone swarms completed more than 20,000 autonomous flights. These micro-drones gave stage performances at New York's Madison Square Garden.

How a drone swarm system works

A drone swarm system has different algorithms. Theoretical frameworks are needed to design distributed flocking algorithms. Several factors need to be considered while designing a drone swarm. Four areas that must be looked at to unlock the full potential of a drone swarm are:

- Swarm size
- Customisation
- Diversity
- Hardening

The larger the size of the swarm, the greater the distance it can cover. For example, large under-water swarms can cover greater distances in search of adversary submarines or surface vessels.

Future drone swarms may not

contain the same type and size of drones—these may consist of small to large-sized drones with different payloads. These may also have drones that are equipped with different sensors to handle different functions. As a result, under-sea drones will be able to communicate with surface and aerial drones.

Robotic researchers have gone one step ahead and introduced drones that can self-organise and exhibit flocking behaviour (like birds). Hungarian Academy of Science has designed a drone swarm that can be operated in the real world without the use of a central control room. They have demonstrated this new model many times using computer simulations.

Constrained motions, communication capabilities, delays, feelings of getting upset or nervous, and so on are some of the factors that are still difficult to achieve while designing drone swarms. Scientists are working to overcome such problems as delays, uncertainties and kinematic constraints.

Each drone in the flock is separately programmed to follow specific flight-path. One of the main things required here is that all drones in the flock fly in

the same direction and follow the same average position.

During flight, drones should not crash with obstacles like trees or collide with each other. However, autonomous collisions and object avoidance are yet unsolved problems. Present models exhibit heavy noise and come with inaccurate sensors and short-range communication devices. These problems need sophisticated analysis.

Swarming drones communicate with each other while in flight and can respond to changing conditions autonomously. A good example of this is a dense flock of singing birds reacting to a sudden threat like a bird of prey. The entire flock then exhibits a series of tasks and pretend to be a single organism.

Underlying challenges in the defence for drone swarms

In future, drone swarms can play a critical role in defence systems and warfare. The drone swarm attack on Russian forces in Syria is a good example of how this technology can be deployed. During the attack, six drones were intercepted by electronic warfare units; three exploded on touching the ground while the other three made landing outside the base. The remaining seven were destroyed using anti-aircraft missiles. This attack reveals that the age of drone swarms has arrived on the warfront.

Military drone swarms may incorporate drones equipped with anti-radiation and anti-jamming weapons. These can block hypersonic missiles. Swarms can be equipped with chemical, biological, radiological and nuclear detectors to provide homeland security.

Applying artificial intelligence (AI) to a drone swarm enables drones to imitate certain animals that work together. This is particularly helpful for synchronised tasks. The swarm's intelligence relies heavily on AI. Real-time communication is important among drones in a swarm as drones need to communicate with each other using an airborne mesh network and with the control centre. A drone swarm can be operated from kilometres away.

“Terrorists and militants can oper-